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Project Presentation3 (370s)

Agile Team Charter

1 Team Purpose

Objectives of the Team

The team's goal is to create WaterWatch, an intelligent water monitoring system that addresses the issue of the invisibility of real-time water use in Cape Town households. The team is tasked not only with designing a product, but also with solving a researched and real problem that affects urban communities in water-stricken areas.

The project is motivated by research that indicates most households lack the ability to monitor their daily water consumption, which can result in hidden leaks, high bills and wastage. As a result the team's challenge is to address this issue through a user-focused solution.

Vision Statement

The team's goal is to design a system that allows households to make meaningful choices about water use through easy-to-access, real-time data. This in turn will help achieve sustainable water consumption and behaviour change at a community level.

Definition of Success

> The team aims to demonstrate a strong and coherent connection between the identified problem and the proposed solution by showing how the WaterWatch system effectively addresses the lack of real-time water usage data in Cape Town households. This is supported by thorough research, including market surveys and analysis, which validate both the need for the solution and the demand within the target community. In addition, the team focuses on delivering a practical, viable and context-sensitive prototype that is technically feasible and grounded in current technology rather than unrealistic assumptions. The project ensures co-inherence across all deliverables, where the research, design and final solution are logically aligned and consistently reinforce one another.

2. Team Roles and Responsibilities

Full Names	Agile Role	Responsibilities
Matimu Mabunda	Project Manager	Facilitates agreed processes; coordinates team's work; tracks progress with task boards; resolves issues blocking progress; maintains communication and collaboration among team members.

Kamvelihle Mqadi	Research Analyst	Performs primary and secondary research; identifies and prioritises product and user requirements; ensures the solution meets the user's needs; maintains product backlog; tests ideas with data.
Marco Julian Human	Frontend Developer & UI Designer	Develops the mobile application's user interface; ensures the user interface is accessible and user-friendly; translates designs into interactive prototypes; works with the back-end developer to integrate the user interface.
Thapelo Masingi	Backend Developer	Manages the technical design and logic of the application; defines how data will be acquired, processed and stored; develops the integration of the smart sensor with the mobile application; ensures data throughput, security and performance; assists in implementing functionality such as real-time data monitoring, alerts and data processing.

3. Team Values and Principles

Core Values

Accountability – Each team member takes full ownership of their assigned tasks and deadlines, ensuring that all work is completed to a high standard. This also includes being honest about challenges, delays, or mistakes and communicating them openly to the team.

Transparency – Open and clear communication is always maintained regarding progress, expectations, and challenges. Tools such as the Kanban board ensure that all work is visible, reducing confusion and preventing hidden or unclear tasks.

Respect – The team values diverse perspectives, skills, and contributions. Respect is demonstrated through active listening, constructive feedback, and maintaining professional communication throughout the project.

Reliability – Team members are expected to consistently meet deadlines and deliver quality work. Being reliable ensures smooth workflow and proper coordination between dependent tasks, such as frontend and backend integration.

Continuous Improvement – The team regularly reflects on its performance and uses feedback to improve both processes and outcomes. Through ongoing retrospectives, areas for improvement are identified and adjustments are made to enhance efficiency and collaboration.

Commitment to Quality – All deliverables are produced to meet both academic and professional standards, ensuring that the final output is of high quality.

Application of Agile Principles

Incremental Delivery – The project is divided into smaller, manageable tasks that can be completed and reviewed regularly, allowing for steady progress and reduced complexity.

Continuous Feedback Integration – Feedback is actively encouraged during team discussions and reviews, and is used to refine ideas, improve designs, and address issues early in the process.

Adaptability and Responsiveness – The team remains flexible and open to change, adjusting plans when new insights emerge from research or when challenges arise during development.

Collaborative Working Approach – Team members work closely together, regularly sharing updates, supporting one another, and making decisions collectively to ensure alignment and knowledge sharing.

User-Centred Focus – All development decisions are guided by the needs of the end user. Features such as real-time monitoring and leak detection are designed specifically to address the lack of water usage awareness identified during research.

4. Communication Plan

Agile Framework

The team uses an Agile logbook template to support an iterative and continuous workflow that aligns with the evolving nature of the project. This logbook serves as a tool for organising and tracking tasks throughout the development process. All work items are categorised into key stages, namely the Product Backlog, In Progress, Review, and Completed, allowing the team to monitor progress, manage priorities, and ensure tasks move efficiently through each phase.

Communication and Task Coordination Platform

The team uses WhatsApp to communicate and discuss between the members about coordination and updates about the tasks. This allows for real-time communication, quick resolution of problems and ongoing engagement with the team throughout the project. Gather in campus to discuss and help each other to see the outcome of project.

Frequency of Meetings

Ceremony	Frequency	Purpose
Stand-up	15 minutes weekdays	Track progress and address issues
Planning	Weekly	Set tasks and priorities
Review	Weekends	Evaluate completed work
Retrospective	Monday and Friday	Evaluate teamwork and processes

Working Agreements

Team members are expected to be prepared and punctual for all meetings, ensuring that time is used effectively. Team members should communicate in advance if we are unable to meet deadlines and actively participate in discussions to contribute to team progress. It is also important to respect different working styles within the team while maintaining a collaborative environment. Additionally,

the team focuses on finding solutions rather than dwelling on problems, promoting a positive and productive workflow.

5. Goals and Milestones

SMART Goals

- **Specific:** Design a smart water monitoring system for Cape Town households that tracks real-time water usage and detects leaks.
- **Measurable:** Successfully complete and submit all deliverables including research report, system design, and interactive prototype.
- **Achievable:** Use available tools and skills within the team to complete the system design and prototype without physical hardware.
- **Relevant:** Address water conservation challenges by improving user awareness of water consumption.
- **Time-bound:** Complete the full project within 9 weeks, following a structured development timeline

Project Milestones

Milestone	Timeline
Research and Validation of Problem	Week 2
Product Concept and Requirements Defined	Week 4
Design and prototype	Week 6
Review and Refinement of Deliverables	Week 8
Final Submission	Week 9